Project: - 1

Title: - Data Leverager – Power Query Data Transformation Project

Sources used: -

Sales_Jan.xlsx Sales_Feb.xlsx Latest Covid-19 India Status.csv Employee_Data.xlsx Sales_Mar.xlsx

Transformation applied: -

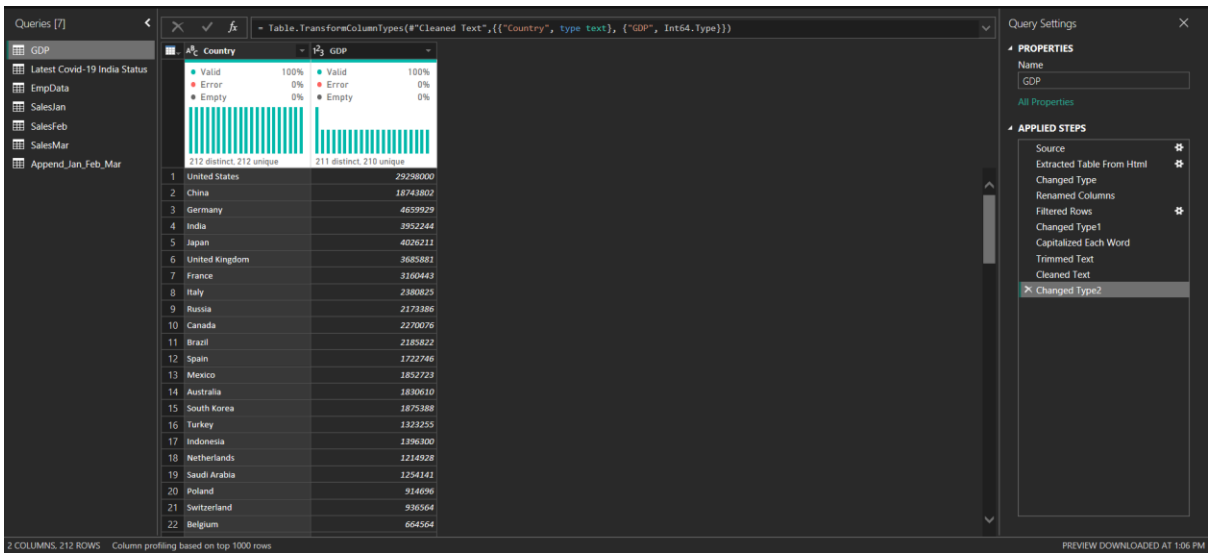
1) First row to header
2) Rename columns according to a data
3) Changes data type according to data
4) Remove null value
5) Apply TRIM(), CLEAN(), SPLIT COLUMN BY DELE,ITER
6) Round off revenue 2 decimal places
7) Create new columns profit using revenue – cost
8) Extract Day, Month, Year
9) Add a calculated age column using Birthdate
10) Create a new column Sales Category
    a. High >= 10000
    b. Medium between 5000 to 9999
    c. Low < 5000
11) Add Index columns
    a. Based on 0
    b. Based on 1
12) Pivot sales data to convert monthly columns into a single column
13) Unpivot columns back into normalized from
14) Merge sales data with employee data using Region or EmployeeID
15) Append Jan, Fed, Mar sales data using Queries as new table
16) Group by Region
    a. Total Sales
    b. Average Order Value
    c. Transaction Count

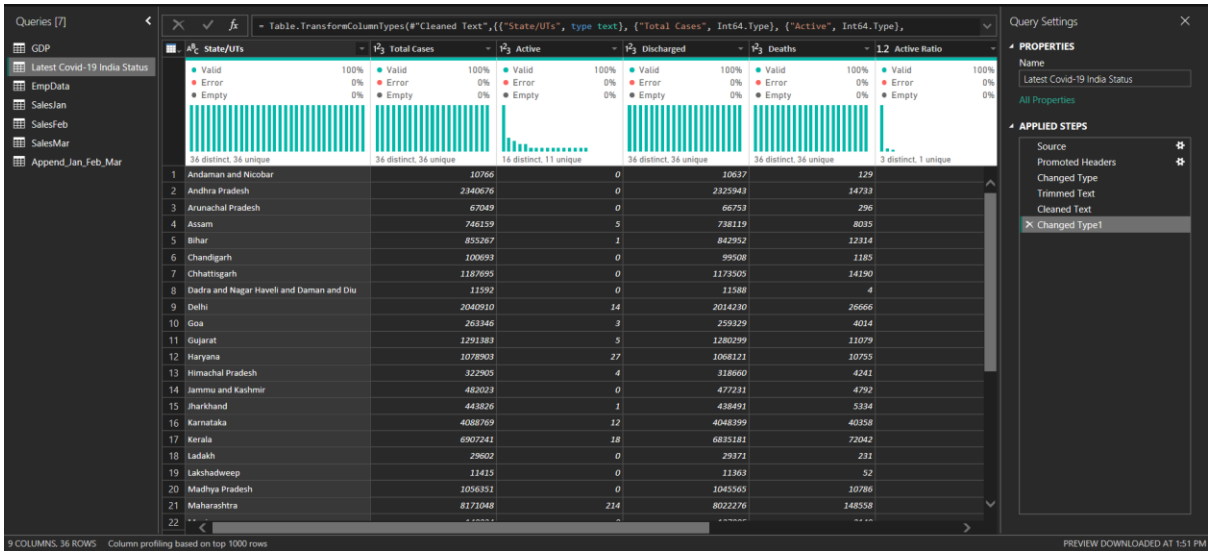
17) Data Profiling and Quality

   a. Identity missing values
   b. Detect column-level errors
   c. Understand distinct and unique values

18) Configure dynamic folder path using Parameters

19) Modify data source credentials under Data Source Settings.





Queries [7]

- GDP
- Latest Covid-19 India Status
- EmpData
- SalesJan
- SalesFeb
- SalesMar
- Append_Jan_Feb_Mar

```
= Table.ExpandTableColumn(#"Merged Queries2", "SalesMar", {"Region", "Total_Sales", "Avg_Order_Value", "Transaction_Counts"},
```

	EmployeeID	First_Name	Last_name	Department	Region	Join_Date	SalesJan.
	6 distinct, 4 unique	6 distinct, 4 unique	6 distinct, 4 unique	4 distinct, 1 unique	4 distinct, 2 unique	6 distinct, 4 unique	5 distinct, 2
1	EMP001	John	Smith	Sales	North	1/15/2020	ORD001
2	EMP001	John	Smith	Sales	North	1/15/2020	ORD001
3	EMP001	John	Smith	Sales	North	1/15/2020	ORD005
4	EMP001	John	Smith	Sales	North	1/15/2020	ORD005
5	EMP005	Chris	Evans	Support	North	2/12/2022	ORD001
6	EMP005	Chris	Evans	Support	North	2/12/2022	ORD001
7	EMP005	Chris	Evans	Support	North	2/12/2022	ORD005
8	EMP005	Chris	Evans	Support	North	2/12/2022	ORD005
9	EMP002	Alice	Brown	Marketing	West	3/10/2019	ORD002
10	EMP003	David	Lee	Operations	South	6/25/2021	ORD003
11	EMP006	Kevin	Hart	Operations	South	11/5/2021	ORD003
12	EMP004	Emma	Wilson	Sales	East	9/30/2018	ORD004

Query Settings

PROPERTIES

Name: EmpData

All Properties

APPLIED STEPS

- Source
- Navigation
- Changed Type
- Capitalized Each Word
- Trimmed Text
- Cleaned Text
- Changed Type1
- Split Column by Delimiter
- Changed Type2
- Renamed Columns
- Uppercased Text
- Merged Queries
- Expanded SalesJan
- Merged Queries1
- Expanded SalesFeb
- Merged Queries2
- Expanded SalesMar

29 COLUMNS, 12 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 10:50 AM

Queries [7]

- GDP
- Latest Covid-19 India Status
- EmpData
- SalesJan
- SalesFeb
- SalesMar
- Append_Jan_Feb_Mar

```
= Table.RenameColumns(#"Reordered Columns3",{{"Value", "Revenue"}})
```

	OrderID	First_Name	Last_Name	Region	Product_Category	Revenue	Cost
	5 distinct, 5 unique	5 distinct, 5 unique	5 distinct, 5 unique	4 distinct, 3 unique	4 distinct, 3 unique	5 distinct, 5 unique	5 distinct, 5
1	ORD006	John	Smith	North	Electronics	11000	
2	ORD007	Alice	Brown	West	Clothing	6000	
3	ORD008	Kevin	Hart	South	Furniture	8000	
4	ORD009	Emma	Wilson	East	Electronics	14000	
5	ORD010	Chris	Evans	North	Books	4000	

Query Settings

PROPERTIES

Name: SalesFeb

All Properties

APPLIED STEPS

- Source
- Navigation
- Changed Type
- Capitalized Each Word
- Trimmed Text
- Cleaned Text
- Changed Type1
- Split Column by Delimiter
- Changed Type2
- Renamed Columns
- Uppercased Text
- Rounded Off
- Added Custom
- Reordered Columns
- Changed Type3
- Reordered Columns1
- Inserted Day Name
- Inserted Month
- Inserted Year
- Inserted Age
- Inserted Total Years
- Inserted Round Down
- Added Conditional Column
- Reordered Columns2

19 COLUMNS, 5 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 1:38 PM

Queries [7]

- GDP
- Latest Covid-19 India Status
- EmpData
- SalesJan
- SalesFeb
- SalesMar
- Append_Jan_Feb_Mar

```
= Table.TransformColumns(#"Calculated Total Years",{{"Birthdate", Number.RoundDown, Int64.Type}})
```

	OrderID	Customer_Name	Region	Product_Category	Revenue	Cost	Order_Da
	5 distinct, 5 unique	5 distinct, 5 unique	4 distinct, 3 unique	4 distinct, 3 unique	5 distinct, 5 unique	5 distinct, 5 unique	5 distinct, 5
1	ORD001	John Smith	North	Electronics	12000	8000	
2	ORD002	Alice Brown	West	Clothing	4500	3000	
3	ORD003	David Lee	South	Furniture	7000	5000	
4	ORD004	Emma Wilson	East	Electronics	15000	9000	
5	ORD005	Chris Evans	North	Books	3000	1500	

Query Settings

PROPERTIES

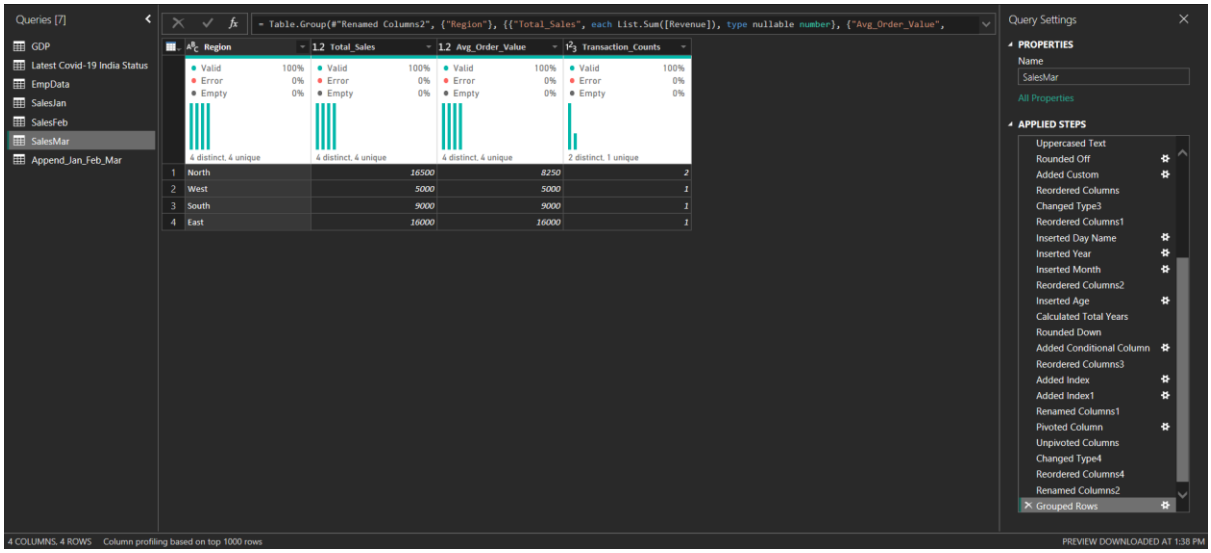
Name: SalesJan

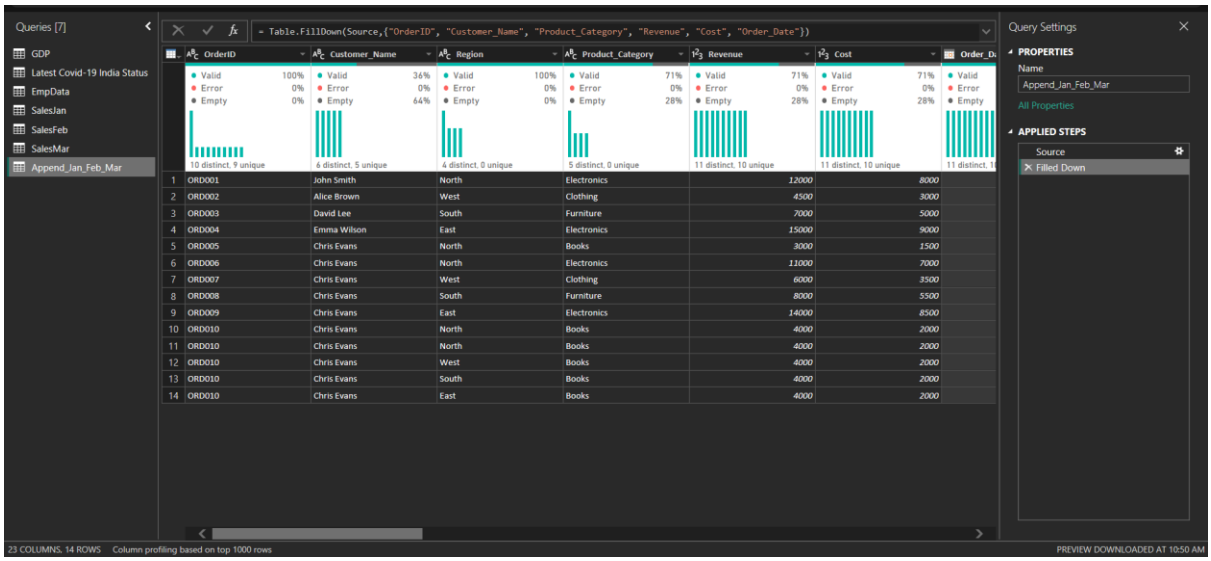
All Properties

APPLIED STEPS

- Source
- Navigation
- Changed Type
- Capitalized Each Word
- Trimmed Text
- Cleaned Text
- Rounded Off
- Changed Type1
- Calculated Age
- Calculated Total Years
- Rounded Down

8 COLUMNS, 5 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 10:50 AM


Queries [7]
GDP
Latest Covid-19 India Status
EmpData
SalesJan
SalesFeb
SalesMar
Append_Jan_Feb_Mar
= Table.Group(#"Renamed Columns2", {"Region"}, {{"Total_Sales", each List.Sum([Revenue]), type nullable number}, {"Avg_Order_Value",
Region
Total_Sales
Avg_Order_Value
Transaction_Counts
Valid 100%
Error 0%
Empty 0%
4 distinct, 4 unique
4 distinct, 4 unique
4 distinct, 4 unique
2 distinct, 1 unique
1 North 16500 8250 2
2 West 5000 5000 1
3 South 9000 9000 1
4 East 16000 16000 1
Query Settings
PROPERTIES
Name
SalesMar
All Properties
APPLIED STEPS
Uppercased Text
Rounded Off
Added Custom
Reordered Columns
Changed Type3
Reordered Columns1
Inserted Day Name
Inserted Year
Inserted Month
Reordered Columns2
Inserted Age
Calculated Total Years
Rounded Down
Added Conditional Column
Reordered Columns3
Added Index
Added Index1
Renamed Columns1
Pivoted Column
Unpivoted Columns
Changed Type4
Reordered Columns4
Renamed Columns2
Grouped Rows
4 COLUMNS, 4 ROWS Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 1:38 PM


Queries [7]
GDP
Latest Covid-19 India Status
EmpData
SalesJan
SalesFeb
SalesMar
Append_Jan_Feb_Mar
= Table.FillDown(Source,{"OrderID", "Customer_Name", "Product_Category", "Revenue", "Cost", "Order_Date"})
OrderID
Customer_Name
Region
Product_Category
Revenue
Cost
Order_D
Valid 100%
Error 0%
Empty 0%
Valid 36%
Error 0%
Empty 64%
Valid 100%
Error 0%
Empty 0%
Valid 71%
Error 0%
Empty 28%
Valid 71%
Error 0%
Empty 28%
Valid 71%
Error 0%
Empty 28%
Valid
Error
Empty
10 distinct, 9 unique
6 distinct, 5 unique
4 distinct, 0 unique
5 distinct, 0 unique
11 distinct, 10 unique
11 distinct, 10 unique
11 distinct, 1
1 ORD001 John Smith North Electronics 12000 8000
2 ORD002 Alice Brown West Clothing 4500 3000
3 ORD003 David Lee South Furniture 7000 5000
4 ORD004 Emma Wilson East Electronics 15000 9000
5 ORD005 Chris Evans North Books 3000 1500
6 ORD006 Chris Evans North Electronics 11000 7000
7 ORD007 Chris Evans West Clothing 6000 3500
8 ORD008 Chris Evans South Furniture 8000 5500
9 ORD009 Chris Evans East Electronics 14000 8500
10 ORD010 Chris Evans North Books 4000 2000
11 ORD010 Chris Evans North Books 4000 2000
12 ORD010 Chris Evans West Books 4000 2000
13 ORD010 Chris Evans South Books 4000 2000
14 ORD010 Chris Evans East Books 4000 2000
Query Settings
PROPERTIES
Name
Append_Jan_Feb_Mar
All Properties
APPLIED STEPS
Source
Filled Down
23 COLUMNS, 14 ROWS Column profiling based on top 1000 rows
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